

PROGRAM 1: FACTORIAL OF A NUMBER

Code:

```
fact=function(num){  
  if (num==0||num==1){  
    return(1)  
  }else{  
    return(num*fact(num-1))  
  }  
  
  cat("Enter a Number:")  
  num<-readLines(con="stdin",n=1)  
  num=as.numeric(num)  
  print(num)  
  result=fact(num)  
  cat("Factorial:",result,' ')
```

OUTPUT (Input: 5):

```
[1] 5  
Factorial: 120
```

PROGRAM 2: PALINDROME CHECKER

Code:

```
palin<-function(result){  
  og<-result  
  rev<-0  
  while(result>0){  
    a<-result%%10  
    rev<-(rev*10)+a  
  
    if(og==rev){  
      cat("It is a pallindrome\n")  
    }else{cat("It is not a pallindrome\n")}  
  }  
  
  cat("Enter a Number:")  
  num<-readLines(con="stdin",n=1)  
  num=as.numeric(num)  
  print(num)  
  palin(num)
```

OUTPUT (Input: 121):

```
[1] 121  
It is a pallindrome
```

PROGRAM 3: LINEAR REGRESSION

Code:

```
data(mtcars)
model=lm(mpg~wt,data=mtcars)
print(summary(model))

newcars<-data.frame(wt=c(0.2,0.3,0.5))
predicted=predict(model,newdata=newcars)
results=data.frame(weight1000lbs=newcars$wt,) Tj 0 -11 Td ( PREDICTED_MG=predicted)
print(results)

plot(mtcars$wt,mtcars$mpg,) Tj 0 -11 Td ( main="MPG VS WEIGHT",xlab="MPG",) Tj 0 -11 Td ( ylab="WEIGHT",pch=
abline(model,col="red",lwd=2)
```

OUTPUT:

```
Call: lm(formula = mpg ~ wt, data = mtcars)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.2851      1.8776  19.858  < 2e-16
wt           -5.3445      0.5591  -9.559  1.29e-10
R-squared:  0.7528, Adjusted R-sq: 0.7446
Predictions: weight1000lbs | PREDICTED_MG
1  0.2 | 38.017,  2  0.3 | 36.677,  3  0.5 | 33.996
[Plot: MPG vs WEIGHT with regression line]
```

PROGRAM 4: LOGISTIC REGRESSION

Code:

```
data<-read.csv("student_data.csv")
print(data)
model<-glm(Result ~ Hours,data=data,) Tj 0 -11 Td (          family=binomial)

cat("Logistic Regression Model Summary:\n")
print(summary(model))
new_data<-data.frame(Hours=5.5)

probability<-predict(model,new_data,) Tj 0 -11 Td (          type="response")
cat("\n Predicted Probability:",probability)
prediction<-ifelse(probability>=0.5,1,0)
cat("Predicted Class(0=Fail,1=Pass):",) Tj 0 -11 Td (    prediction,"\n")
class<-ifelse(prediction==1,"Pass","Fail")
cat("Final Prediction:",class,"\n")
```

OUTPUT:

```
Call: glm(formula = Result ~ Hours, family = binomial)
Coefficients: (Intercept)=-4.0777, Hours=0.8240
Residual deviance: 38.23 on 18 df, AIC: 42.23
```

```
Predicted Probability of passing: 0.7342
Predicted Class: 1
Final Prediction: Pass
```